

No. of Functional Features Included in the Solution

Date	1 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – AI-Based Crop Recommendation System
Maximum Marks	5 Marks

Functional Features Overview:

This document describes the functional features implemented in the **OptiCrop – AI-Based Crop Recommendation System**. Each feature contributes to recommending the most suitable crop based on soil nutrients and environmental conditions using Machine Learning and provides users with an easy-to-use web interface for entering agricultural parameters and viewing crop recommendation results.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Crop Recommendation Form	Allows users to enter soil nutrient values and parameters for prediction.	Frontend	Done	High
2	Data Preprocessing	Cleans and prepares input data for prediction.	Machine Learning	Done	High
3	Machine Learning Model	Predicts the most suitable crop using the ML model.	ML Model	Done	High
4	Crop Recommendation Result	Displays the crop based on the user's input.	Flask Backend	Done	High
5	Responsive User Interface	Provides a responsive, and user-friendly interface.	Frontend	Done	High
6	Input Validation	Validates user input before prediction.	Backend	Done	Medium
7	Model Integration	Integrates the trained Machine Learning model with the Flask application.	Flask Backend	Done	Medium
8	GitHub Deployment	Stores and manages project source code.	Version Control	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home Page, About Page, Crop Recommendation Form
2	Backend / Logic	3	Data Preprocessing, Input Validation, Crop Prediction Processing
3	Database / Storage	1	Crop Recommendation Dataset (CSV), Trained Model (model.pkl) & Scaler (scaler.pkl)
4	API / Integration	1	Flask Integration with Machine Learning Model
5	Security / Authentication	1	Input Validation and Secure Data Processing